

Mode 2 — Baseline-BFP

DP16 (8×8) → DP16 (8×4) → DP8 (8×4) → block exponents

Step 1 — DP16 (8×8)

$$\text{DP16 } (8 \times 8) = \sum_{i=0}^{15} a_i b_i$$

Step 2 — DP16 (8×4)

$$\text{DP16 } (8 \times 8) = \underbrace{2^4 \sum_{i=0}^{15} a_i b_i^{hi}}_{\text{DP16 } (8 \times 4)} + \underbrace{\sum_{i=0}^{15} a_i b_i^{lo}}_{\text{DP16 } (8 \times 4)}$$

Step 3 — DP8 (8×4)

$$\text{DP16 } (8 \times 8) = \underbrace{2^4 \sum_{i=0}^7 a_i b_i^{hi}}_{\text{DP8 } (8 \times 4)} + \underbrace{2^4 \sum_{i=8}^{15} a_i b_i^{hi}}_{\text{DP8 } (8 \times 4)} + \underbrace{\sum_{i=0}^7 a_i b_i^{lo}}_{\text{DP8 } (8 \times 4)} + \underbrace{\sum_{i=8}^{15} a_i b_i^{lo}}_{\text{DP8 } (8 \times 4)}$$

Step 4 — Block exponents (one exponent block per DP8 (8×4))

$$\left(2^4 \sum_{i=0}^7 a_i b_i^{hi} + \sum_{i=0}^7 a_i b_i^{lo}, E_0 \right), \left(2^4 \sum_{i=8}^{15} a_i b_i^{hi} + \sum_{i=8}^{15} a_i b_i^{lo}, E_1 \right) \rightarrow (M, E)$$

$$E_k = E_{a,k} + E_{b,k}; \quad E = \max(E_0, E_1), \quad \delta_k = E - E_k; \quad \text{truncating shifts, on block sums only}; \quad \text{DP16 } (8 \times 8) = M \cdot 2^E.$$

Step 5 — Result

$$M = 2^{-\delta_0} \left(2^4 \sum_{i=0}^7 a_i b_i^{hi} + \sum_{i=0}^7 a_i b_i^{lo} \right) + 2^{-\delta_1} \left(2^4 \sum_{i=8}^{15} a_i b_i^{hi} + \sum_{i=8}^{15} a_i b_i^{lo} \right), \quad E = \max(E_0, E_1)$$